

Product Requirements Document (PRD)

Zero Downtime AI

AI-Powered Industrial Continuity & Predictive Automation System

1. Executive Summary

Project Name

Zero Downtime AI

Tagline

Predict. Prevent. Perform.

Project Type

Industrial AI + Automation + Predictive Maintenance Platform

Target Event

ABB Accelerator Hackathon

Theme Alignment

Primary Theme:

- Next-Gen Control System Interface

Secondary Alignment:

- Industrial Automation
 - AI-Powered Operations
 - Smart Manufacturing
 - Industry 4.0
 - Predictive Systems
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2. Problem Statement

Industrial environments suffer major operational and financial losses due to unexpected machine failures and downtime.

Traditional maintenance systems are mostly reactive:

- Machines fail first
- Engineers respond later
- Downtime already occurs
- Production efficiency drops
- Maintenance costs increase

Current challenges include:

- Lack of predictive intelligence
- Delayed fault detection
- High operational downtime
- Manual monitoring systems
- Inefficient maintenance scheduling
- No automated preventive action systems

Industries require a smarter system capable of:

- Monitoring machine health continuously
 - Predicting failures before breakdowns
 - Automating preventive responses
 - Assisting operators in real time
 - Reducing downtime significantly
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3. Proposed Solution

Zero Downtime AI is an AI-powered industrial monitoring and predictive automation platform designed to prevent machine downtime before failures occur.

The platform uses:

- Simulated industrial sensor data
- AI-based anomaly detection
- Predictive failure analysis
- Smart automation workflows
- Real-time monitoring dashboards
- Automated preventive response systems

The system continuously analyzes machine health parameters such as:

- Temperature
- Vibration
- Power consumption
- Pressure
- RPM

When abnormal behavior is detected: 1. AI predicts possible failure 2. Risk score is generated 3. Alerts are triggered 4. Preventive automation actions are executed 5. Operators are notified 6. Downtime is minimized or prevented

4. Vision Statement

To build an intelligent industrial continuity system capable of transforming traditional reactive maintenance into proactive AI-driven operational automation.

5. Objectives

Primary Objectives

- Reduce unexpected industrial downtime
- Predict failures before breakdowns
- Automate preventive operational responses
- Improve industrial efficiency
- Demonstrate Industry 4.0 concepts
- Showcase AI-driven automation

Secondary Objectives

- Create visually impressive industrial dashboards
 - Build scalable architecture
 - Deliver a hackathon-ready prototype
 - Simulate real industrial workflows
 - Demonstrate business impact
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6. Target Users

Primary Users

- Factory Operators
- Plant Managers
- Maintenance Engineers
- Industrial Supervisors

Secondary Users

- Smart Manufacturing Companies
 - Industrial Automation Providers
 - Operational Technology Teams
 - Energy Optimization Teams
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7. Core Features

Feature 1 — Real-Time Industrial Dashboard

Description

A centralized dashboard displaying machine health, analytics, alerts, and AI predictions.

Functionalities

- Machine health cards
- Live sensor metrics
- Real-time status monitoring
- Failure probability indicators
- AI risk scoring
- Alert notifications
- Machine analytics
- Operational summaries

Dashboard Components

Top Navigation

- Project branding
- Factory status indicator
- Notification center

- System uptime status

Machine Cards

Each machine card displays:

- Machine ID
- Current status
- Temperature
- Vibration level
- Power consumption
- Failure risk
- Last maintenance timestamp

Live Analytics Charts

Charts required:

- Temperature graph
- Power consumption graph
- Vibration analysis graph
- Failure probability trend
- Operational efficiency chart

UI Requirements

- Dark industrial UI
 - Modern responsive design
 - Clean visual hierarchy
 - Smooth animations
 - Real-time updates
 - Industrial control panel aesthetics
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Feature 2 — Sensor Simulation Engine

Description

A simulated industrial sensor environment generating realistic machine operational data.

Simulated Parameters

- Temperature
- Vibration
- Voltage
- Current

- Pressure
- RPM
- Load percentage
- Power usage

Requirements

- Real-time updates every 2–5 seconds
- Randomized but realistic values
- Configurable thresholds
- Multiple machine support
- Abnormality injection capability

Abnormal Event Simulation

The system should simulate:

- Overheating
 - Excessive vibration
 - Power spikes
 - Pressure anomalies
 - Sudden load increase
 - Efficiency degradation
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Feature 3 — AI Prediction Engine

Description

An AI module responsible for predicting machine failures using anomaly detection and predictive analytics.

AI Goals

- Detect abnormal machine behavior
- Predict possible failures
- Generate risk scores
- Trigger preventive workflows

Suggested ML Models

Preferred:

- Isolation Forest
- Random Forest
- Logistic Regression

Optional:

- LSTM (if advanced implementation possible)

Input Parameters

- Temperature
- Vibration
- Voltage fluctuations
- RPM variation
- Historical operational patterns

Outputs

- Machine health score
- Failure probability percentage
- Predicted failure time
- Risk category
- Suggested action

Risk Categories

- Healthy
- Moderate Risk
- High Risk
- Critical

Example Output

Failure Probability: 84% Predicted Failure Time: 2 Hours Suggested Action: Reduce load and inspect cooling system.

Feature 4 — Smart Alert System

Description

An intelligent industrial alert engine that notifies operators about risks and abnormalities.

Alert Types

- Warning alerts
- Critical alerts
- Maintenance alerts
- System anomaly alerts

- Preventive action notifications

Alert Channels

- Dashboard notifications
- Email notifications (optional)
- Push notifications (optional)

Example Alerts

- Machine 4 overheating
- Bearing failure predicted
- Power instability detected
- Downtime risk identified

Alert Prioritization

Priority levels:

- Low
 - Medium
 - High
 - Critical
-

Feature 5 — Automation Response Engine

Description

A preventive automation system capable of taking intelligent actions when anomalies are detected.

Purpose

This is the core innovation feature.

Instead of only detecting failures, the system also demonstrates automated preventive action.

Automated Responses

- Reduce machine load
- Trigger maintenance scheduling
- Notify operators
- Activate backup workflow
- Log anomaly events

- Suggest operational optimization

Example Workflow

1. AI detects overheating
2. Risk score becomes critical
3. System reduces machine load
4. Maintenance ticket generated
5. Operator notified
6. Downtime prevented

Workflow Engine Requirements

- Rule-based automation logic
 - Event-driven architecture
 - Trigger-response system
 - Expandable automation modules
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Feature 6 — Maintenance Recommendation System

Description

Provides AI-generated maintenance recommendations.

Suggested Recommendations

- Cooling system inspection
 - Bearing replacement
 - Load balancing
 - Motor calibration
 - Voltage stabilization
 - Preventive shutdown scheduling
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Feature 7 — Historical Analytics

Description

Store and analyze historical operational data.

Functionalities

- Historical sensor logs
- Downtime analysis
- Risk trend analysis
- Maintenance history
- AI prediction history

Analytics Goals

- Identify recurring issues
 - Improve prediction accuracy
 - Demonstrate long-term intelligence
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8. Optional Advanced Features

Voice Assistant

Example Commands

- Show critical machines
- Display machine health
- Open risk dashboard
- Show downtime statistics

Technologies

- Web Speech API
 - SpeechRecognition
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Digital Twin Visualization

Description

Visual simulation of industrial machines and their live operational state.

Goal

Increase presentation quality and Industry 4.0 alignment.

Mobile Responsive Support

Dashboard should support:

- Desktop
 - Tablet
 - Mobile view
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AI Chat Assistant

Optional AI assistant capable of:

- Explaining anomalies
 - Suggesting maintenance
 - Answering operator queries
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9. Functional Requirements

FR-1

The system shall display real-time machine health data.

FR-2

The system shall simulate industrial sensor data.

FR-3

The AI engine shall detect anomalies.

FR-4

The system shall predict possible failures.

FR-5

The system shall generate risk scores.

FR-6

The system shall trigger alerts.

FR-7

The system shall automate preventive responses.

FR-8

The system shall log historical operational data.

FR-9

The dashboard shall update dynamically.

FR-10

The system shall support multiple machines.

10. Non-Functional Requirements

Performance

- Dashboard updates under 1 second
- Real-time simulation performance
- Low latency alerts

Scalability

- Modular architecture
- Expandable machine support
- API-based backend

Reliability

- Stable simulation engine
- Error handling support
- Fallback UI states

Usability

- Clean modern UI
- Beginner-friendly navigation
- Professional industrial design

Security

- Basic authentication (optional)
- Protected APIs
- Environment variable management

11. System Architecture

High-Level Architecture

Sensor Simulation Layer ↓ Data Processing Layer ↓ AI Prediction Engine ↓ Risk Analysis System ↓
Automation Response Engine ↓ Dashboard + Alerts ↓ Historical Analytics

Frontend Architecture

Frontend/ ├── Dashboard ├── Components | ├── MachineCards | ├── Charts | ├── Alerts
| ├── RiskPanel | ├── Navbar ├── Pages ├── Services ├── Hooks ├── Context ├──
Utilities

Backend Architecture

Backend/ ├── API ├── Controllers ├── Routes ├── Services ├── SimulationEngine ├──
PredictionEngine ├── AutomationEngine ├── AlertSystem ├── Database

AI Architecture

AI/ ├── Dataset ├── ModelTraining ├── Prediction ├── RiskScoring ├── Utilities

12. Recommended Tech Stack

Frontend

- React
- Tailwind CSS
- Chart.js or Recharts
- Framer Motion

Backend

- Node.js
- Express.js

AI/ML

- Python
- Scikit-learn
- Pandas
- NumPy

Database

Preferred:

- Firebase

Alternative:

- MongoDB

Hosting

Frontend:

- Vercel

Backend:

- Render or Railway
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13. API Requirements

Machine APIs

GET /machines Returns machine list.

GET /machine/:id Returns machine details.

GET /machine/:id/metrics Returns live metrics.

Prediction APIs

POST /predict Returns prediction result.

Response Example: { "risk": "High", "probability": 84, "predictedFailureTime": "2 hours", "suggestedAction": "Reduce machine load" }

Alert APIs

GET /alerts Returns all active alerts.

POST /alerts Creates new alert.

Analytics APIs

GET /analytics/history Returns historical operational data.

14. Data Simulation Logic

Healthy State

Temperature: 50–70°C Vibration: Low Power Usage: Stable

Warning State

Temperature: 70–85°C Vibration: Moderate Power Fluctuations: Minor

Critical State

Temperature: >90°C Vibration: High Power Spikes: Severe

15. AI Workflow

1. Sensor data generated
 2. Data processed
 3. AI model evaluates anomalies
 4. Risk score calculated
 5. Failure probability generated
 6. Alerts triggered
 7. Automation engine activated
 8. Dashboard updated
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16. User Flow

Operator Workflow

1. Open dashboard
 2. View machine health
 3. Monitor alerts
 4. Observe AI predictions
 5. Review preventive actions
 6. Confirm maintenance recommendations
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17. UI/UX Guidelines

Design Direction

Industrial futuristic aesthetic.

Color Scheme

- Dark background
- Neon warning colors
- Green healthy indicators
- Red critical alerts

UI Goals

- Highly visual
- Dashboard-first design
- Smooth transitions
- Professional industrial appearance

Inspiration

ABB industrial dashboards SCADA systems Modern manufacturing control panels Tesla industrial UI style

18. Demo Workflow

Demo Storyline

Stage 1 — Normal Operations

All machines healthy. Dashboard green.

Stage 2 — Sensor Anomaly

Machine temperature increases.

Stage 3 — AI Detection

AI identifies abnormal behavior.

Stage 4 — Prediction

Failure probability generated.

Stage 5 — Automation Trigger

System automatically:

- reduces machine load
- schedules maintenance
- sends alert

Stage 6 — Result

Downtime prevented successfully.

19. Hackathon Submission Deliverables

Mandatory Deliverables

1. Working Prototype

Requirements:

- Functional dashboard
- AI simulation
- Alert system
- Automation workflow

2. Presentation PPT

Slides required:

- Problem Statement
- Solution
- Architecture
- Features
- AI Workflow
- Demo Screenshots
- Future Scope
- Impact

3. Demo Video

Recommended Duration: 2–4 minutes

Video Structure

- Team introduction
- Problem explanation
- System walkthrough
- Live demo
- Future scope

4. GitHub Repository

Must include:

- README
- Architecture diagram
- Setup guide
- Screenshots

- Demo instructions
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22. Success Metrics

Technical Metrics

- Real-time dashboard updates
- Successful anomaly detection
- Automated workflow execution
- Stable UI performance

Presentation Metrics

- Professional appearance
- Clear workflow demonstration
- Strong storytelling
- Industrial relevance

Hackathon Metrics

- Innovation quality
 - Practical implementation
 - ABB alignment
 - Demo effectiveness
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23. Future Scope

Future Enhancements

- Real IoT sensor integration
- Cloud deployment
- Advanced deep learning models
- Digital twin systems
- Edge AI deployment
- Predictive energy optimization
- Multi-factory monitoring
- Autonomous industrial orchestration

Enterprise Possibilities

- Smart manufacturing platform
- Industrial SaaS product

- Factory optimization suite
 - Industrial AI assistant
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24. Final Positioning Statement

Zero Downtime AI is an intelligent industrial continuity platform that combines predictive maintenance, anomaly detection, and automation workflows to prevent machine downtime before failures occur.

The system demonstrates Industry 4.0 principles through:

- AI-driven operational intelligence
- Real-time industrial analytics
- Automated preventive responses
- Smart factory concepts
- Predictive automation systems

The platform is designed to showcase how AI can transform traditional industrial maintenance into proactive intelligent operations.

25. Suggested Final Submission Tagline

Zero Downtime AI Predict. Prevent. Perform.

AI-Powered Industrial Continuity & Predictive Automation System.